Getting Started With 2D Apps: Porting Android Apps to Meta Quest

START Boot Camp Session 3

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Date: December 2024

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Introduction

This guide explains how to develop and optimize 2D apps for the Meta Horizon Store and platform. It covers key features supported on Meta Quest, the native 2D app development process, and the progressive web app (PWA) integration with the browser. The guide also introduces the Meta Horizon Store, with insights on publishing and monitoring your apps.

Learning Objectives

By the end of this guide, you will:

- Learn how to port an Android app (sample or personal) to a 2D panel.
- Understand how users interact with 2D panel apps and the limitations of traditional inputs and interactions.

2D apps on Meta Horizon

The Meta Horizon Store reached a major milestone in 2024: it is now open for all developers to publish 2D apps. While apps like WhatsApp and Messenger have been available for some time, this update creates new opportunities for mobile app developers to explore mixed reality (MR). Developers can adapt existing apps or create new ones for this platform.

Key Advantages for Mobile Developers:

- Built on Android: Meta Horizon OS is built on the Android Open Source Project (AOSP), ensuring compatibility with most Android APIs. This simplifies the process of transitioning Android apps to Horizon.
- Core Functionality Support: Meta Horizon OS and its Shell handle essential functions like app positioning, input management, and audio/video playback. This makes the platform function like an innovative Android device.
- Familiar Development Environment: Android and web developers will find the development environment intuitive, making it easier to bring native 2D apps or PWAs to the platform.

Platform Features

The Meta Horizon platform enhances the functionality and user experience of 2D apps in mixed reality. With adaptive panel sizing, versatile input handling, advanced audio management, and media playback options, Meta Horizon OS brings your apps to life in immersive environments.

Panels

Panels solve the challenges of adapting phone apps to mixed reality (MR) by offering flexible sizing and multitasking capabilities.

- **Customizable Panel Sizing:** Resize 2D apps freely to any orientation or use preset modes for convenience.
- **Seamless Multitasking:** Run and interact with multiple 2D apps simultaneously without pausing or leaving immersive experiences.
- **Enhanced Spatial Control:** Use overlay mode to keep 2D apps open while engaging with immersive games and experiences.

Input

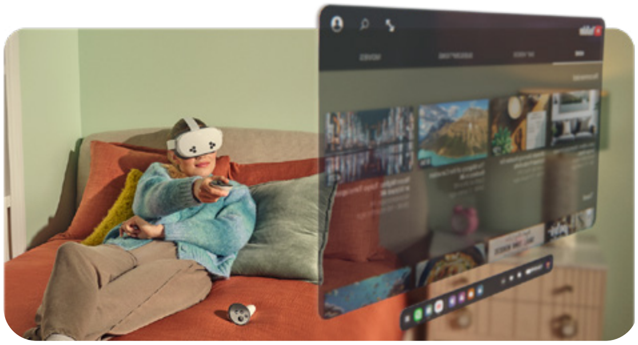
Meta Horizon simplifies input management, ensuring seamless integration across various input devices.

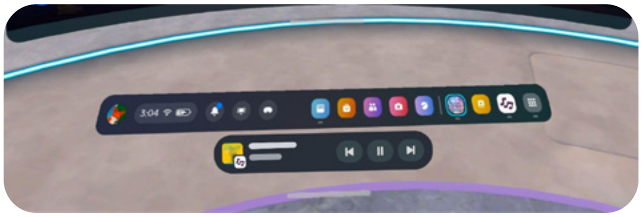
- **Unified Input Handling:** The Shell and underlying services automatically route inputs from controllers, hand gestures, and Bluetooth peripherals to the active app. Developers don't need to configure additional input settings.
- **Virtual Keyboard:** Includes features like swipe typing, suggestions, and autocomplete, making text entry intuitive and efficient across apps.

Audio Features

Meta Horizon OS elevates the audio experience in mixed reality with continuous playback and immersive audio enhancements.

- **Background Audio:** Continue listening to music from minimized apps with dedicated media playback controls. Start or pause music directly from the media playback bar when apps are minimized.
- **Volume Mixer:** Adjust call volume and app or media volume separately, allowing tailored audio settings for different streams.
- **Spatial Audio:** Create realistic audio experiences by spatializing sounds from 2D apps without requiring extra coding.
- **Dolby Atmos Support:** Enjoy enhanced audio quality for supported content, starting with streaming services like Netflix in the browser.





Media Features

Media consumption on Meta Horizon OS is transformed through intuitive and immersive features, enhancing the experience of 2D apps while adding a layer of interactivity to your app's content.

- **Theater View:** Switch any media app into Focus Mode with a single click, featuring a large, customizable screen for an immersive experience.
- **Customizable Viewing Experience:** Switch any media app into Focus Mode with a single click, featuring a large, customizable screen for an immersive experience.

Travel Mode

Travel Mode extends the utility of Meta Quest, enabling users to engage with apps during flights or offline scenarios.

- **Offline on Inflight Functionality:** Play downloaded media during flights or connect to in-flight entertainment systems.
- **Private Viewing:** Whether traveling by air or train, enjoy content on a personal large screen that delivers enhanced privacy and comfort.

Travel mode is available on trains and planes.

Horizon OS software updates

Meta Horizon OS is constantly evolving, with new features released nearly every month. These updates are designed to improve the platform and enhance the development experience for 2D apps. The goal is to make app development for Meta Horizon OS seamless and enjoyable.

May
v65

- Spatial videos and panoramas
- Travel mode

June
v66

- Background audio
- Media control bar

July
v67

- Swipe typing
- Immersive media continuity
- Theater view
- Movable panels for multitasking

August
v68

- Improved performance for 2D apps
- Controller pairing in-headset
- Audio balancing
- Keyboard improvements

September
v69

- Seamless multitasking
- Spatial panel audio
- Window layout
- Bluetooth quick pairing
- Stylus support

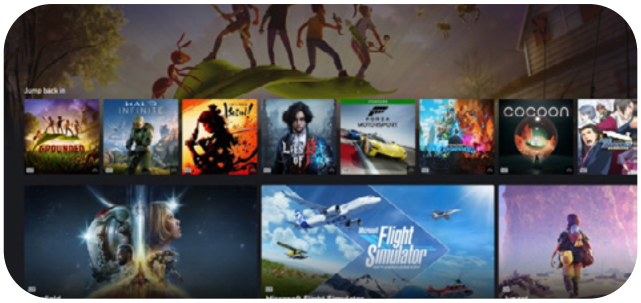
Developer Journeys

Meta Horizon OS offers tailored paths for Android and web developers to expand their reach. Android apps can be adapted and published on the Meta Horizon Store, leveraging existing frameworks to reach new users. Web developers can convert their apps into Progressive Web Apps (PWAs), enhancing functionality with Meta Horizon OS features while increasing visibility. Both approaches provide efficient ways to introduce 2D apps to a growing audience.

Android 2D

Seamlessly port existing mobile Android or create new Android apps to Meta Horizon OS. No barrier for used frameworks.

Ideal for native Android developers.



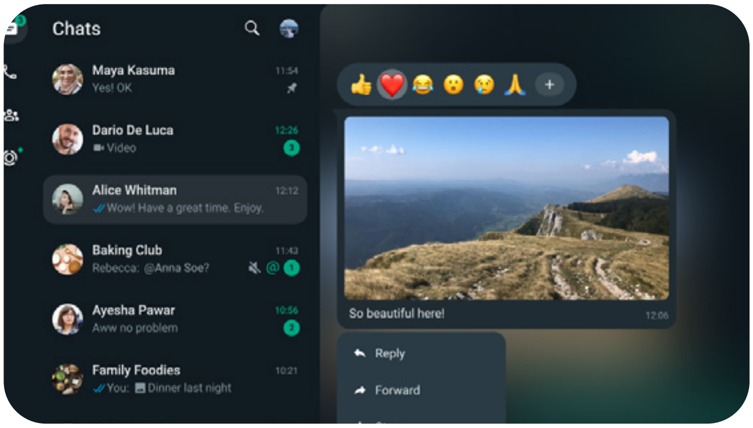
More devices

Web app packaged as a progressive web app (PWA), launched as a standalone self-contained 2D app.

Ideal for web developers.

Android 2D

Meta Horizon OS now supports seamless integration of Android apps, enabling developers to bring their existing mobile applications effortlessly to the Meta Quest. Built on the Android Open Source Project (AOSP), the platform ensures broad compatibility with most Android APIs, allowing apps to operate smoothly and reliably.



Messenger and WhatsApp are already live.
Facebook and Instagram recently launched.

Porting Guide for Android 2D

Follow these steps to port your existing Android mobile apps to Meta Quest and ensure a successful transition to the platform:

1. **App setup on Meta Quest**
   - Continue using your existing tools and development environment. No additional plugins or extensions are required.
   - Enable developer mode on your headset and add it as a target device to build and run your app.
   - Debug and profile your app using the same tools, with breakpoints and stack traces functioning as they would on other Android devices.

2. **Configure App to cross-platform**
   - Ensure cross-platform compatibility for easier future maintenance.
   - Use product flavors as the preferred Android approach.
   - This allows you to manage one codebase with multiple platform targets and manifests to configure specific requirements for each target.

3. **Optimize and enhance**
   - Integrate additional APIs from Horizon OS, such as DRM enabling or panel sizing, to take full advantage of platform capabilities.
   - Review user permissions to ensure smooth operation during the porting process.

4. **Publish your app**
   - Set up a developer account and create an app listing on the Meta Horizon Store using the Meta Quest Developer Dashboard.
   - Upload your APK via the Meta Quest Developer Hub or OVR command line tools (CLI).
   - Submit your app for review, select a launch date, and deploy it to the Meta Horizon Store.

Enhancements

Explore the APIs and configuration options Horizon OS offers to optimize your 2D app's functionality and user experience.

Media protection

The platform ensures robust media protection with features like DRM support and enhanced privacy controls:

- **DRM Support:** Supports Widevine L1 and L3 DRM for secure playback of DRM-protected media. Unauthorized screen capturing or casting results in a blacked-out screen, protecting sensitive content.
- **Enhanced Privacy with Inhibit Capture:** Use the inhibit_capture flag to automatically black out app screens during recording or casting, safeguarding private content such as login credentials or financial information.

```
<!-- To block capturing of DRM content -->
<meta-data
   android:value="secure" <!-- or inhibit_capture -->
   android:name="com.oculus.vrshell.uses_secure_layer" />
```

Panel size customizations
Developers have greater control over app panel sizes to ensure an optimal user experience:

- **Customizable Panel Size:** Adjust app panel sizes to optimize the user experience.
- **Control Over Resizing:** By default, users can resize apps freely, but resizing can be restricted using the supports-free-resizing tag.
- **Aspect Ratio Lock:** Lock the aspect ratio to maintain a consistent user experience and avoid unsupported dimensions disrupting the interface.
- **Set Size Limits:** Implement size boundaries to prevent users from enlarging or reducing the interface in ways that compromise usability.
- **Default Launch Size:** Define the default size for app launch using the layout tag.

```xml
<!-- To enable or disable users from app panel resizing -->
<meta-data
   android:name="com.oculus.vrshell.supports_free_resizing"
   android:value="false" />

<!-- To lock aspect ratio -->
<meta-data
   android:name="com.oculus.vrshell.free_resizing_lock_aspect_ratio"
   android:value="true"/>

<!-- Set app panel size resizing limits [minWidth, maxWidth, minHeight, maxHeight]-->
<meta-data
   android:name="com.oculus.vrshell.free_resizing_limits"
   android:value="800, 1600, 345, 1000" />
```

Refinements

Meta Horizon OS also includes refinements to usability, platform compatibility, and performance, helping developers deliver high-quality apps.

Usability

- **Increase Tap Targets:** Ensure interactive elements are at least 40px in size to make tapping easier and more accessible for users.
- **Add Padding:** Add extra padding around smaller UI components, such as checkboxes, to improve accessibility.

Device/platform limitations

- **Functionality Restrictions:** Adapt your app for the platform's limitations, such as the lack of cellular calling and contacts functionality.
- **Review User Permissions:** Familiarize yourself with the restricted user permissions list to align the app's permissions with platform constraints.
- **Performance Considerations:** With multitasking and resource sharing across simultaneously running apps, ensure smooth operation through profiling and performance testing.

Third-party services

- **Unsupported Services:** Google mobile services, such as Google Identity, push notifications, and in-app purchases via the Google Play Store, are not supported.

Android platform SDK

The Android platform SDK equips developers with essential tools to integrate Meta Horizon-specific features and functionalities into their Android apps, ensuring seamless monetization and compliance.

- **Billing Integration:** Use the Meta Horizon Billing Compatibility SDK to integrate the Billing API into your app. This allows direct monetization via the Meta Quest Store or a smooth transition from the Google Play Billing library.
- **Support for All IAP Types:** Supports consumable items, durable goods, and subscription-based in-app purchases (IAP), offering flexible monetization strategies.
- **User-Age Category API:** Incorporate user-age categorization to align with age-related requirements, especially for apps targeting or including users under 13.

```
implementation("com.meta.horizon.platform.ovr:android-platform-sdk:71+")
implementation("com.meta.horizon.billingclient.api:horizon-billing-compatibility:1.0.0+")
```

Comprehensive documentation on the Billing Compatibility SDK, including guides and helpful links, is available on the developer documentation site.

Meta Spatial SDK: Beyond Android 2D

The Meta Spatial SDK enables Android developers to move beyond traditional 2D apps by integrating advanced graphics and mixed reality (MR) features. This SDK allows developers to create immersive and hybrid experiences that combine 2D functionality with interactive, mixed-reality elements, delivering engaging applications on the Meta Horizon platform.

- **Immersive Experience Integration:** Use the Spatial SDK to transform your traditional 2D app into an engaging mixed-reality experience.
- **Hybrid App Development:** Develop hybrid apps that combine 2D functionality with 3D and immersive elements.
- **Advanced Graphics and MR Features:** Incorporate cutting-edge graphics and mixed reality technologies to enhance your app's visual and interactive appeal.

Progressive Web Apps & Browser

In addition to Android 2D apps, the Meta Horizon platform fully supports Progressive Web Apps (PWAs) and browser-based experiences. These options provide developers with versatile tools to deliver engaging, app-like functionality directly through the web.

Progressive Web Apps

PWAs transform standard websites into installable, app-like experiences that can be deployed on Meta Quest alongside mobile and desktop platforms.

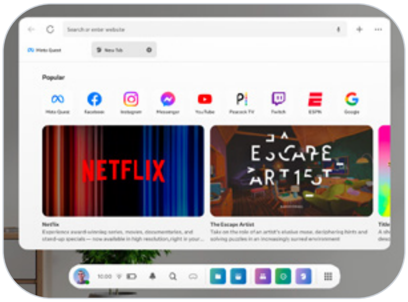
- **System Integration:** PWAs leverage the power of the browser to access Meta Horizon OS features like Theater View for immersive media and Spatial Audio.
- **Additional Discovery and Distribution:** Publishing a PWA to the Meta Quest Marketplace expands your app's reach, offering streamlined discovery for Meta Quest users.

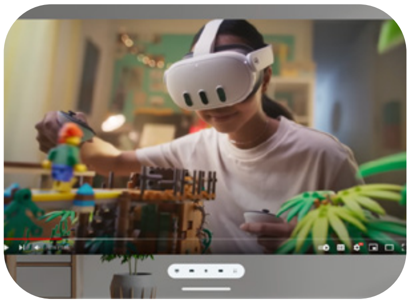
Browser

The browser, powered by the Chromium rendering engine, ensures compatibility with the latest web technologies and delivers exceptional performance on all Meta Quest devices.

- **Comprehensive Media Support:** Supports immersive media playback, including rectilinear, 3D, 180-degree, and 360-degree content, offering an enhanced viewing experience.
- **Enhanced Viewing Features:** Use Theater View and Spatial Audio to make media consumption more immersive. These features make the browser one of Meta Horizon's top-performing 2D apps.

- **Robust Content Protection:** Includes Widevine L1 and L3 DRM support to ensure secure playback of protected content.
- **WebXR Integration:** Supports the WebXR standard, enabling virtual and augmented reality experiences directly within the browser.





Coming Soon: A new way to package PWAs

Meta Horizon is introducing a new approach to packaging Progressive Web Apps (PWAs) by transitioning from the previous CLI method to Bubblewrap. This standardized tool simplifies the creation and distribution of PWAs across platforms, aligning with typical Android processes. Additionally, it integrates in-app purchase capabilities via the Digital Goods API, offering developers streamlined development, improved system integration, and expanded monetization opportunities.

Meta Horizon Store

The Meta Quest Store is the final step in your development journey, whether your app is a PWA, Android 2D, or a spatial app. The store provides crucial visibility and engagement opportunities, connecting your app with the extensive Meta Quest user base.

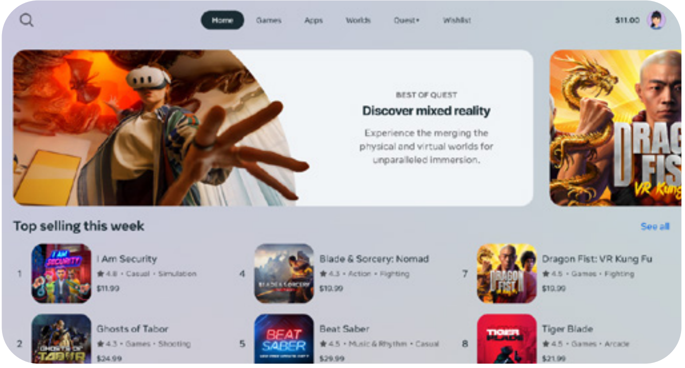
This section outlines the steps to prepare and publish your app effectively.

Publishing your app

To launch your app on the Meta Horizon Store, follow these steps:

Meta Quest Developer Dashboard
Begin by creating a developer account, which enables you to add and manage app listings directly on the dashboard.

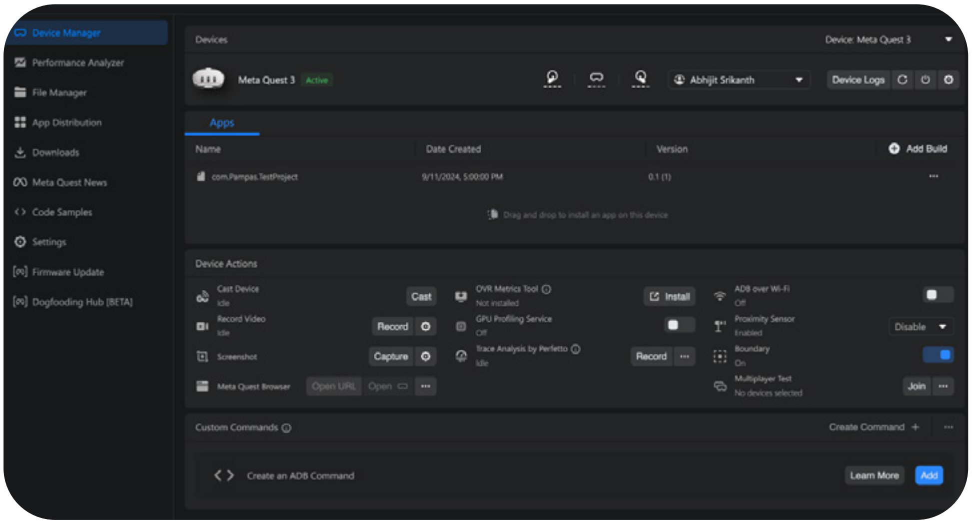
- **Create a Developer Account:** Begin by creating a developer account to manage app listings directly through the dashboard.
- **Add App Metadata:** Fill out key app details, including the app name, descriptions, screenshots, and icons.
- **Complete Verification Steps:** Follow the verification process outlined by the Meta Quest Store to ensure your app meets all publishing requirements.



Meta Quest Developer Hub
The Meta Quest Developer Hub is a standalone desktop app that not only helps you upload your app to the store but also to sideload it onto devices for testing. It includes a performance analyzer to monitor vital metrics like memory usage, CPU load, and FPS.

The CLI tool can also be used to upload binaries or integrate with your Continuous integration systems for automated publishing.

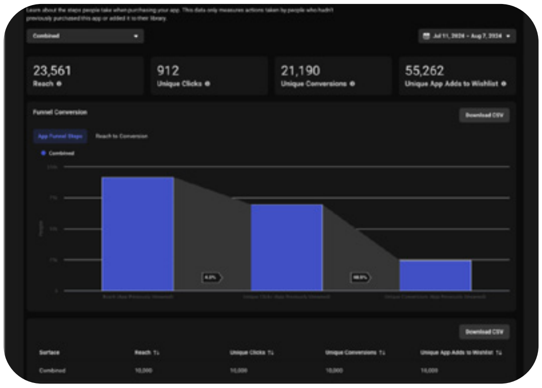
- Upload binaries to app listing
- Developer options to cast, capture in high-definition
- Performance analyzer and OVR metrics

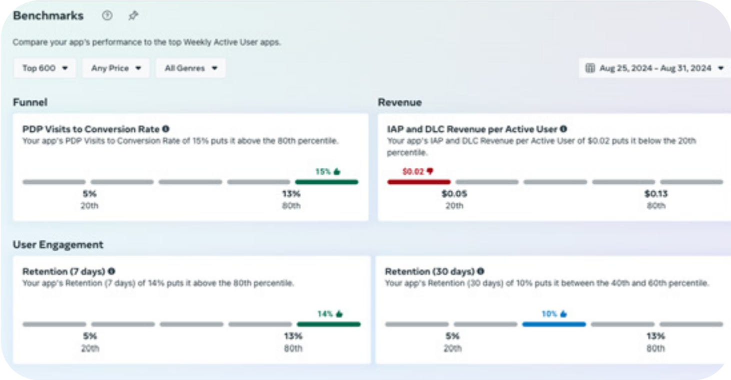


Monitoring your app

The Meta Quest Store dashboard provides a range of tools and features to help you monitor and optimize your app's performance. These insights allow you to better understand your audience and improve user engagement.

- **Analytics:** Monitor key metrics such as installs, active users, time spent, and user reviews to assess your app's performance and how users are engaging with it.
- **Benchmark Metrics:** Use the benchmark metrics tab to compare your app's performance with others on the Meta Quest Store. Identify potential areas for improvement to stay competitive.
- **Engage Your Audience:** Use push notifications and developer posts to connect with your audience. Build relationships by responding to user reviews and fostering a loyal community around your app.
- **Pre-Launch Community Building:** Leverage options like pre-orders, coming-soon pages, and early access releases to attract interest and gather feedback before launch. These tools allow you to create a community that supports your app from the start.





Key Takeaways

Comprehensive Support: Whether working with native Android apps or Progressive Web Apps (PWAs), Meta Horizon OS provides robust support and development tools. Developers can adapt or enhance their apps using familiar tools and processes.

Simplified Publishing: The Meta Quest Store and its developer tools make the publishing process straightforward, from submission to launch.

Opportunities for Growth: With the Meta Horizon Store open to all 2D apps, developers have a clear path to port existing applications or create new ones, reaching broader audiences and gathering valuable insights to improve their apps.

Get started with the 2D App Development Documentation to migrate your app today!